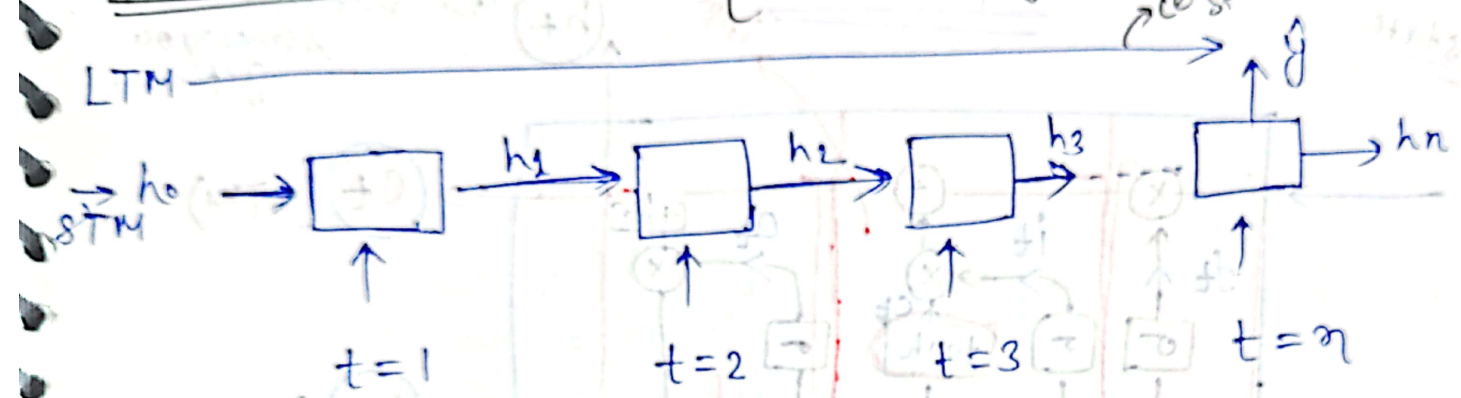


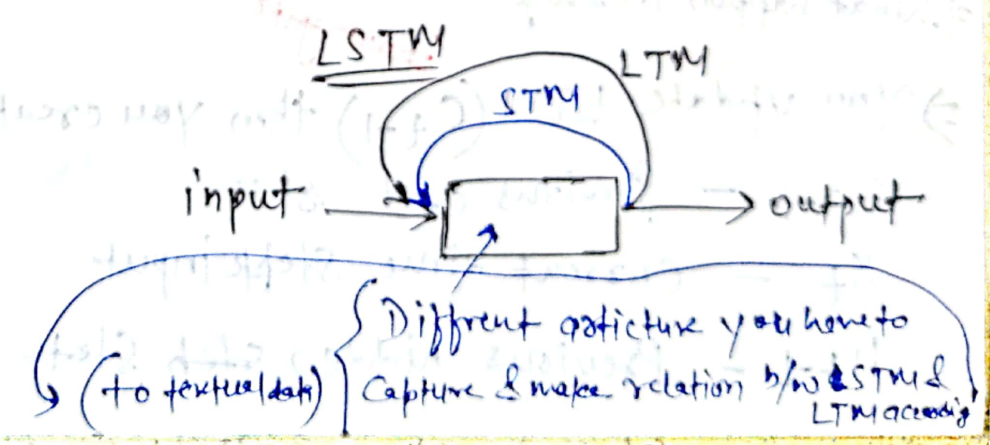
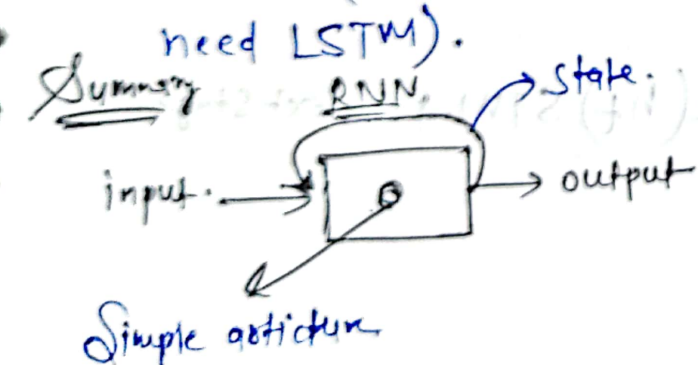
#) Long short term memory (LSTM)

LSTM core idea :-

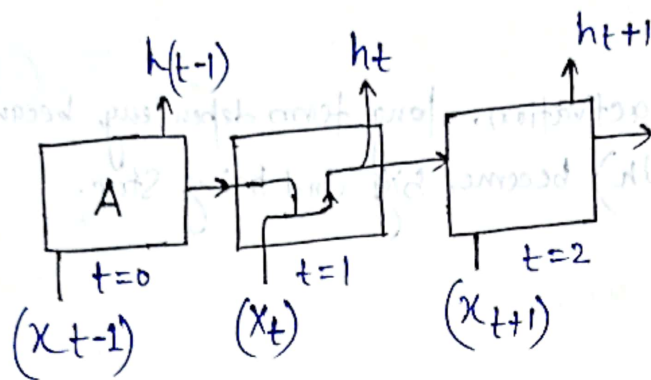
{ STM $\rightarrow$ short term memory
LTM $\rightarrow$ long term memory }



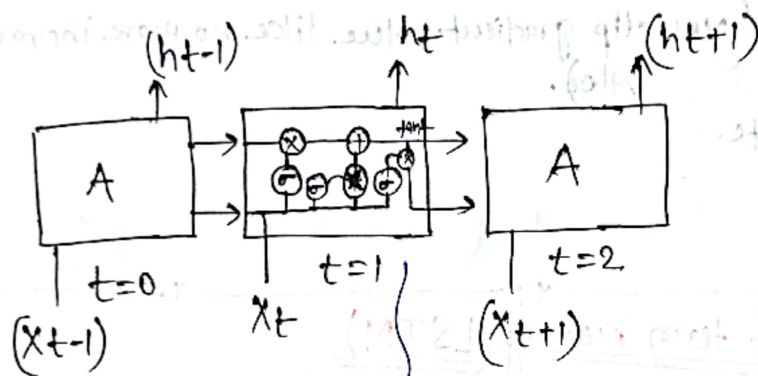
$\Rightarrow$ In LSTM you introduced a path of long term memory & short term memory also. if anything needs important according to LTM you put/sent to LTM and it will be stand at that time where you can't remove by yourself. (In case of RNNs he forget about LTM that's why we



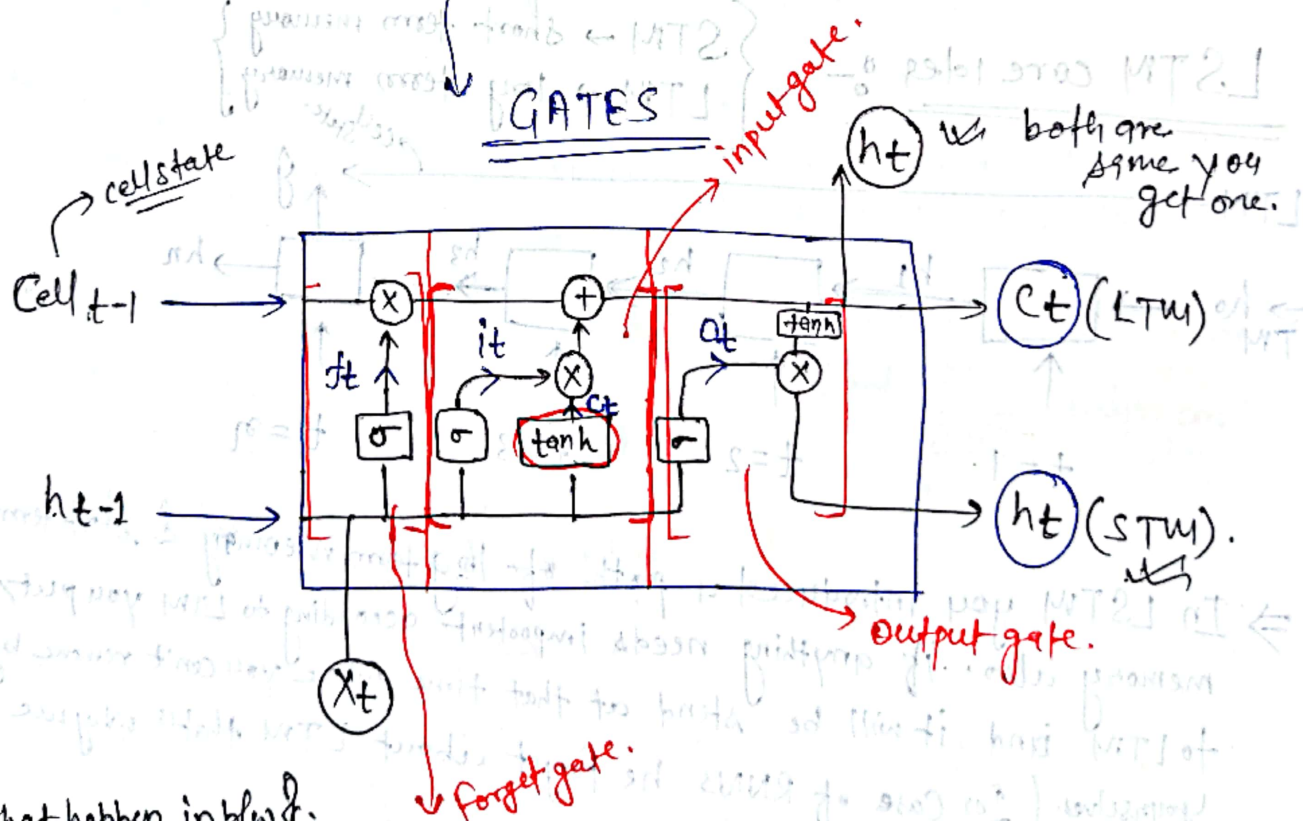
#). Diagram of both RNN & LSTM :-



RNN



LSTM



#). What happens in b/w?

⇒ You update LTM (C_{t-1}) then you create (h_t) STM. for next stage.

C_{t-1} - Previous cell state.

x_t - Current time step input

h_{t-1} - Previous hidden step state.

#/ LSTM Gates

- ① forget Gates :- It decide which part remove from LTM based on current input. (to remove something from c_t)
- ② Input Gates :- It decide what should add in LTM (like new information based on current input. (to add connection to c_t))
- ③ Output Gates :- It decide which one should remove from LTM and show as output to visualize/show based on current input. and it also create STM (Calculate hidden state h_t).

#/ $(h_t \& c_t)$ are equal dimension and both are vectors.

$\Rightarrow (h_t \& c_t)$ dimension never be change.

$\Rightarrow X_t$ is nothing but word to vector converter. you can says kind of t

$\Rightarrow X_t$ is a vector. like Ex:- when we are converting text vectorization using one hot encoding ---

$$\begin{aligned} \text{Cat.} &= [1 \ 0 \ 0] \\ \text{mat} &= [0 \ 1 \ 0] \\ \text{rat} &= [0 \ 0 \ 1] \end{aligned}$$

$$\begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}$$

$f_t \rightarrow$ it found (forget gate)

$i_t \rightarrow$ it found (input gate)

$c_t \rightarrow$ Candidate cell State (it found input gate)

$o_t \rightarrow$ it found in (output gate)

#/ $[f_t, i_t, c_t, o_t]$ are same no. of dimension like (c_t, h_t)

#. Pointwise operations

$\otimes$

$+$

$\tanh$

let $C_{t-1} = [4 \ 5 \ 6]$

$f_t = [1 \ 2 \ 3]$

$C_{t-1} \otimes f_t \rightarrow$ vectors.

$\Rightarrow [4 \ 10 \ 18]$

$+$

Pointwise addition. b/w vectors

$\tanh$

Ex:- $\tanh(4)$. for this vector

$[0.26 \ 0.34 \ 0.53]$

it is also a vector.

#. NN layers :-

σ

Sigmoid.

$\tanh$

also activation.

NN layers. which is containing nodes. has a activation Sigmoid or tanh

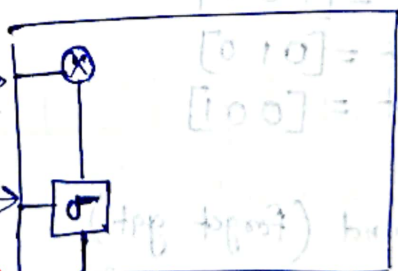
#. Forget Gate :-

① Calculate f_t

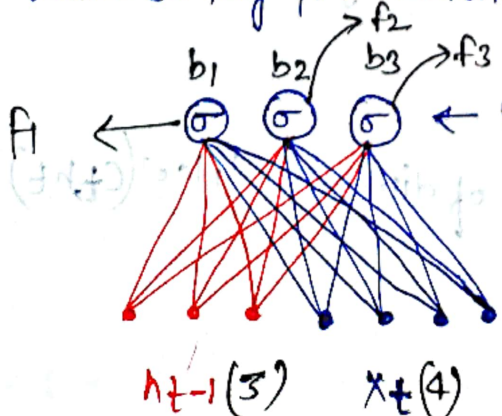
② $C_{t-1} \otimes f_t$

that means you forget or remove something from cell state.

C_{t-1}
 (h_{t-1})
(3dim vector)



x_t (4dim vector)



$\Rightarrow (f_1, f_2, f_3)$ output & dimension is 3

$\Rightarrow$ weights shape $(3 \times 7) (7 \times 1) + (3 \times 1) \text{ bias} = (3 \times 1) f_t$
 $\Rightarrow$ after concatenate we get (21 weights) + 4 bias

$$f_t = \sigma \left(W_f [h_{t-1}, x_t] + b_f \right) \quad (3 \times 1)$$

(3×7) (7×1) (3×1)
 Sigmoid. $\rightarrow$ concatenate

$$[W_f] \cdot [h_{t-1}, x_t] \quad (3 \times 7) \cdot (7 \times 1)$$

$$(1 \times 8) = (1 \times (3 \times 1)) \cdot b_f (3 \times 1)$$

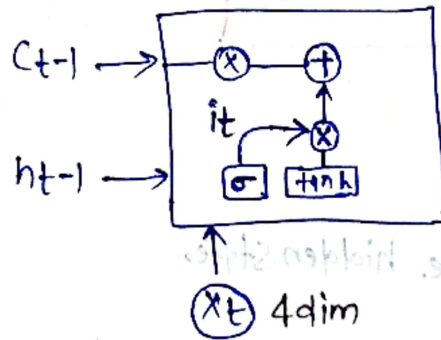
$$f_t = (3 \times 1)$$

$$f_t \otimes C_{t-1} \quad \text{you get vector } (3 \times 1)$$

(3×1) (3×1)
 it represent you get (C_{t-1}) and remove something

Input Gate :-

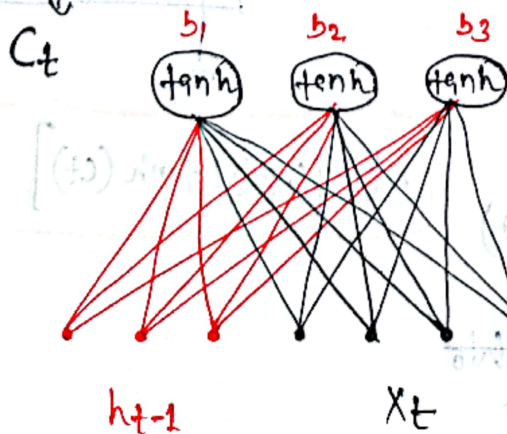
Add some new impⁿ information into input C_t



Stage :-

1. C_t Candidate cell state
2. it decides which value add in cell state C_{t-1}
3. C_t

no. of units in $\tanh$ (3-dim)



$C_t = \text{Potential impotent information}$

$$W_c (3 \times 7)$$

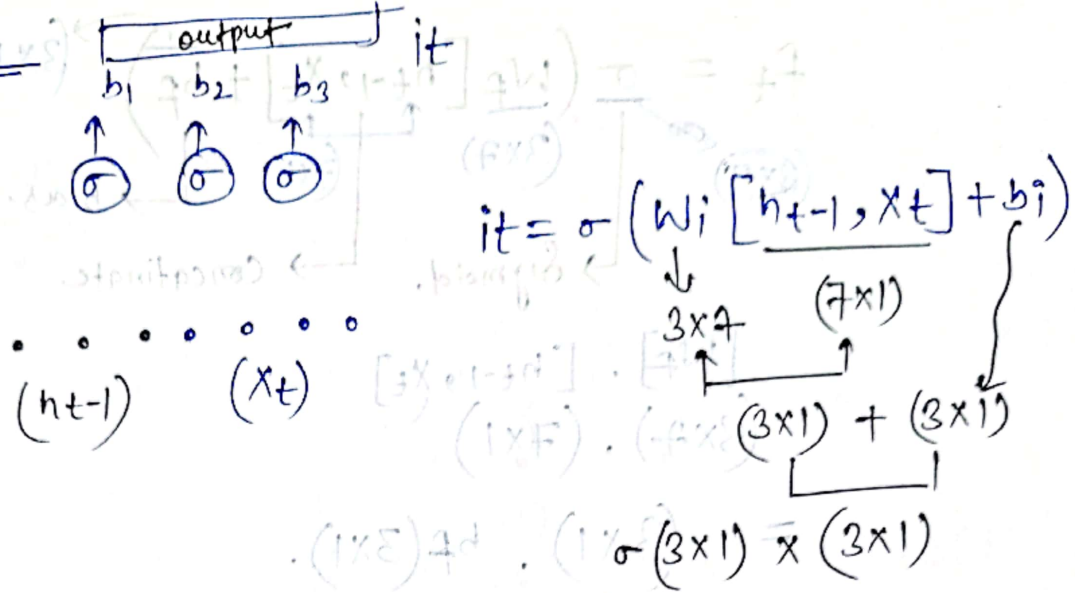
$$C_t = \tanh(W_c [h_{t-1}, x_t] + b_c)$$

(3×7) (7×1)
 $(3 \times 1) + (3 \times 1)$

$$\tanh(3 \times 1) \times (3 \times 1)$$

$$C_t = (3 \times 1)$$

#> for it



$$it = 3 \times 1$$

#> Point wise operation :-

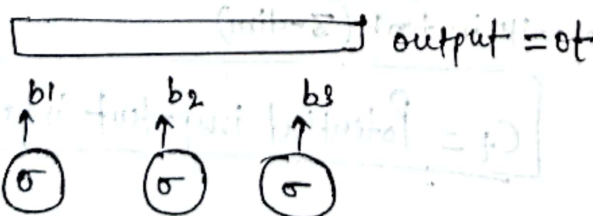
$$it \otimes \bar{c}_t \rightarrow c'_t \quad (\text{filter candidate cell state})$$

Dimensions: $(3 \times 1) \quad (3 \times 1)$

$$c_t = f_t \otimes c_{t-1} \oplus it \otimes c_t \quad \text{Final quantity}$$

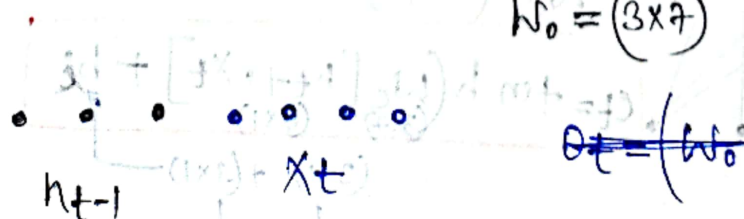
#> Output Gate :-

current time step to decide hidden state.



$$W_o = (3 \times 7)$$

$$h_t = [ot \otimes \tanh(c_t)]$$



$$O_t = \sigma \left(W_o [h_{t-1}, x_t] + b_o \right)$$

$$\begin{matrix} \downarrow & & \downarrow \\ 3 \times 7 & & (7 \times 1) \\ \uparrow & & \uparrow \\ (3 \times 1) & + & (3 \times 1) \end{matrix}$$

$$\sigma = (3 \times 1) \times (3 \times 1)$$

$$= (3 \times 1)$$

$$h_t = O_t \otimes \tanh(C_t)$$

$$\begin{matrix} \downarrow & & \downarrow & & \downarrow \\ (3 \times 1) & & (3 \times 1) & & (3 \times 1) \end{matrix}$$

$h_t = (3 \times 1)$